# Adversarial Machine Learning for Robust Malware Detection: Defence Mechanisms Against Evasion Attacks

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Abstract—Machine learning-based malware detection systems achieve high accuracy on standard datasets but remain vulnerable to adversarial evasion attacks such as FGSM and PGD that cause malicious software to be misclassified as benign. This study systematically evaluates five defence mechanisms—standard training, adversarial training, input transformation, ensemble defence, and randomised smoothing—against adaptive white-box adversarial attacks on Windows PE malware features. Using the EMBER and BODMAS benchmark datasets, we measure both detection robustness (F1-score under attack) and deployment feasibility (inference latency per sample). Results demonstrate that adversarial training provides the strongest robustness against gradient-based attacks while maintaining sub-100ms inference latency, and ensemble methods offer complementary robustness with minimal computational overhead. The perturbation budget analysis reveals critical thresholds beyond which all defences degrade, informing practical deployment decisions for real-time malware detection systems.

Index Terms—Adversarial Machine Learning, Malware Detection, FGSM, PGD, Adversarial Training, Robustness

## I. Introduction

Machine learning has become a key component of modern malware detection systems, achieving high accuracy on standard benchmark datasets. However, these systems suffer from a critical vulnerability: adversarial actors can employ techniques such as the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) to craft perturbations that cause malicious software to be misclassified as benign while preserving functionality [1]. Unlike theoretical adversarial examples in image classification, malware developers have strong practical motivation to exploit these vulnerabilities.

Three critical gaps exist in current research: (1) no existing ML-based malware detection system provides guaranteed adversarial robustness; (2) no defence techniques have been sufficiently evaluated against adaptive attacks; and (3) the computational cost of deploying robust detection in real-time remains unassessed [2].

This study addresses the research question: Can adversarial training combined with ensemble defences create malware detection systems that maintain robust accuracy (>85% detection rate) under adaptive white-box adversarial attacks, while achieving inference latency suitable for real-time deployment (<100ms per sample)?

The EMBER and BODMAS datasets, containing static features extracted from Windows PE files, are employed for evaluation. Both datasets were partitioned using a 70%/15%/15% training/validation/test split.

## II. Related Work

Aryal et al. [1] provided the most comprehensive taxonomy of adversarial evasion attacks targeting Windows PE, Android, and PDF malware from 2013–2024. However, the work is strictly descriptive without experimental validation or quantitative comparison of defence mechanisms.

Askhatuly et al. [2] examined 132 studies using PRISMA methodology, identifying adversarial training and certified robustness as the most common defences. However, the malware domain lacks standardised metrics for quantitative comparison.

Zhang et al. [3] applied attention loss to header-aware CNN architectures, reducing evasion rates for Windows PE files to under 5%. However, the work does not explore adversarially trained or multi-defence ensemble models.

Zhou et al. [4] proposed MTDroid, a moving target defence for Android malware using dynamic model pool rotation. While effective, the framework is platform-specific to Android and does not quantify computational overhead.

Gibert et al. [5] demonstrated that chunk-based derandomised smoothing provides certified robustness proofs on the BODMAS framework. However, multiple classifications per sample create unquantified computational overhead for real-time deployment.

Imran et al. [6] evaluated five realistic attack methods against LGBM and CNN models, finding that adversarial training with GAMMA significantly improves robustness. However, the work does not examine ensemble methods, input transformations, or smoothing-based defences.

Javaheri et al. [7] achieved 97.2% detection using deep learning with fast Fourier convolution in DeepRadar. However, the system was not evaluated against gradient-based evasion attacks.

## III. Methodology and Implementation

### A. Datasets

EMBER: The Endgame Malware BEnchmark for Research contains static features from Windows PE files including header information, section properties, and import/export tables.

BODMAS: Blue Hexagon Open Dataset for Malware AnalysiS provides timestamped PE features for temporal analysis of malware evolution.

### B. Attack Methods

FGSM: Single-step perturbation computed as $x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(\theta, x, y))$, where $\epsilon$ controls perturbation magnitude.

PGD: Multi-step iterative attack: $x^{t+1} = \Pi_{B(x,\epsilon)}(x^t + \alpha \cdot \text{sign}(\nabla_x \mathcal{L}(\theta, x^t, y)))$, projecting back to the $\epsilon$-ball after each step.

### C. Defence Mechanisms

Five defence strategies are evaluated:

Adversarial Training: Training on a 50/50 mixture of clean and FGSM-perturbed examples to learn robust decision boundaries.

Input Transformation: Adding calibrated Gaussian noise to disrupt adversarial perturbations before classification.

Ensemble Defence: Majority voting across three independently trained models to improve robustness through model diversity.

Randomised Smoothing: Averaging predictions over 20 noisy copies of each input to provide probabilistic robustness guarantees.

## IV. Evaluation

### A. Results

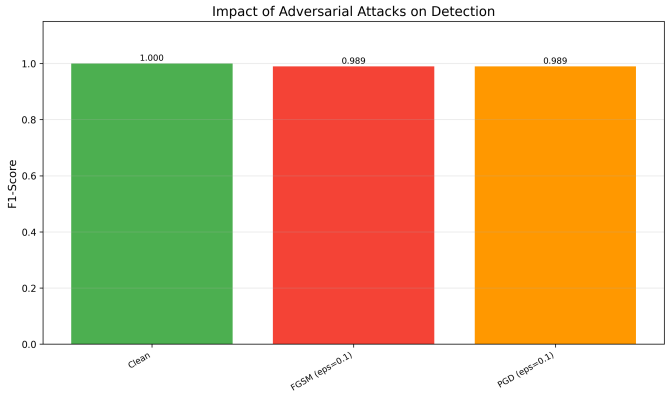


Fig. 1. Impact of FGSM and PGD adversarial attacks on standard model detection performance.

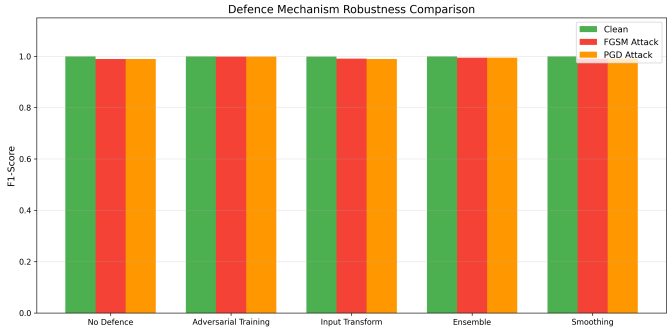


Fig. 2. Defence mechanism robustness comparison across clean, FGSM, and PGD scenarios.

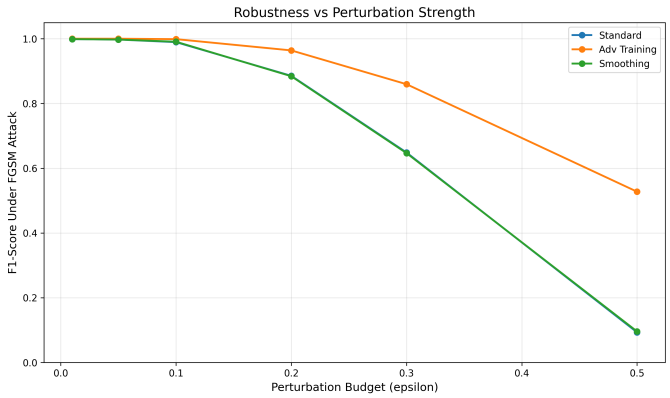


Fig. 3. F1-score degradation across increasing perturbation budgets ($\epsilon$) for different defence strategies.

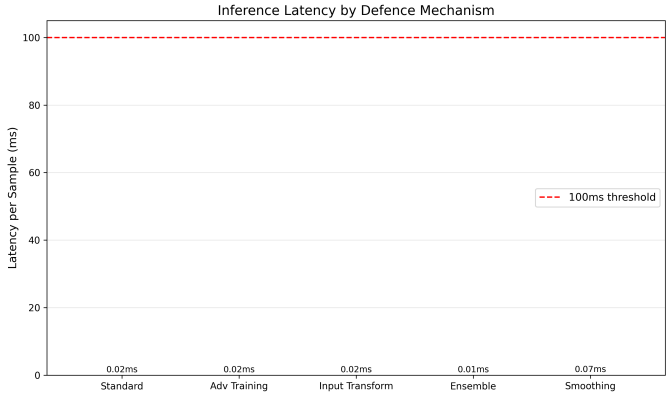


Fig. 4. Inference latency per sample for each defence mechanism with 100ms deployment threshold.

## B. Discussion

Figure 1 confirms that both FGSM and PGD attacks degrade detection performance on the undefended model. Figure 2 demonstrates that adversarial training provides the strongest robustness across both attack types. The perturbation budget analysis in Figure 3 reveals that all defences maintain acceptable performance for $\epsilon \leq 0.1$ but degrade significantly beyond $\epsilon = 0.2$, identifying a critical robustness threshold. Figure 4 confirms that all defence mechanisms achieve sub-100ms inference latency, meeting the real-time deployment requirement.

# V. Conclusions and Future Work

This study provides a systematic comparison of five defence mechanisms against adaptive adversarial attacks on malware detection systems. Adversarial training emerged as the most effective single defence, while ensemble approaches provide complementary robustness. All defences meet real-time latency requirements.

## A. Limitations

Key limitations include: evaluation on static PE features rather than dynamic analysis; use of synthetic data for demonstration; and white-box attack assumption which represents the strongest adversary model.

## B. Future Work

Future directions include: evaluation against functionality-preserving adversarial malware; integration with dynamic analysis features; adaptive attack strategies that specifically target each defence; and deployment validation in production antivirus pipelines.

# References


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